

Spatial Structure of Complex Cell Receptive Fields Measured with Natural Images

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Summary

Neuronal receptive fields (RFs) play crucial roles in visual processing. While the linear RFs of early neurons have been well studied, RFs of cortical complex cells are nonlinear and therefore difficult to characterize, especially in the context of natural stimuli. In this study, we used a nonlinear technique to compute the RFs of complex cells from their responses to natural images. We found that each RF is well described by a small number of subunits, which are oriented, localized, and bandpass. These subunits contribute to neuronal responses in a contrast-dependent, polarity-invariant manner, and they can largely predict the orientation and spatial frequency tuning of the cell. Although the RF structures measured with natural images were similar to those measured with random stimuli, natural images were more effective for driving complex cells, thus facilitating rapid identification of the subunits. The subunit RF model provides a useful basis for understanding cortical processing of natural stimuli.

Introduction

The response properties of neurons in the primary visual cortex have been studied extensively. The spatial structure of cortical RFs was originally examined with relatively simple stimuli, such as oriented bars of light (Hubel and Wiesel, 1962). These studies revealed the clear structure of simple cell RFs with distinct ON and OFF subregions, which directly predicts the orientation and spatial frequency tuning of the cell (Movshon et al., 1978b). Complex cell RFs measured with this technique, however, reveal only the location and overall size of the RF. There is little correspondence between the measured RF structure and the orientation or spatial frequency tuning of the neuron, and such a description of the RF is insufficient for understanding the function of the neuron in visual processing. The difficulty in characterizing complex cell RFs is primarily caused by the nonlinearity of these cells, as reflected in their insensitivity to the luminance polarity of the stimulus (Hubel and Wiesel, 1962). Further studies measuring the nonlinear interaction between paired bars have revealed more detailed spatial structure of complex cell RFs

(Movshon et al., 1978a; Emerson et al., 1987) that can predict the spatial frequency tuning of the cells. The resulting description of complex cell RFs, however, has been limited to the spatial axis perpendicular to the preferred orientation of the cell.

An alternative method of characterizing the RFs of visual neurons is to use large ensembles of complex stimuli, which reduces a priori assumptions about which stimulus features are relevant to the neuron, thus allowing a less biased characterization of the response properties (Touryan and Dan, 2001). The most commonly used complex stimuli are pseudorandom stimuli such as sparse noise (Jones and Palmer, 1987; DeAngelis et al., 1993), white noise (Reid et al., 1997), and dynamic gratings (Ringach et al., 1997; Felsen et al., 2002; Mazer et al., 2002), which have been used successfully in conjunction with linear analyses (e.g., spike-triggered average, referred to as STA) to measure simple cell RFs. Since organisms evolve and develop in the natural environment, natural scenes provide a class of complex stimuli that may be more effective for characterizing neuronal RFs (Rieke et al., 1995). In recent studies, responses of sensory neurons to natural stimuli have been analyzed with modified STA methods (Theunissen et al., 2000, 2001; Smyth et al., 2003) or a recursive least-squares algorithm (Ringach et al., 2002) to estimate the linear RFs of these neurons.

The above techniques, however, are inadequate for estimating the nonlinear RFs of complex cells in the visual cortex. An alternative method, spike-triggered covariance (STC), which computes the eigenvalues and eigenvectors of the second-order Wiener kernel, has been shown to be effective for analyzing the nonlinear RF properties in the fly visual system (de Ruyter Van Steveninck and Bialek, 1988; Brenner et al., 2000). Recently this method has been used to analyze the responses of cortical complex cells to random-bar stimuli aligned to the preferred orientation of the cell (Touryan et al., 2002; Rust et al., 2004). In the present study, we have computed the two-dimensional spatial structure of complex cell RFs from their responses to natural images, using a modified STC analysis that corrects for the spatial correlations in natural images (Field, 1987; Dong and Atick, 1995; Simoncelli and Olshausen, 2001). This analysis reveals the detailed spatial structure of complex cell RFs, which directly corresponds to the orientation and spatial frequency tuning of the cells.

Results

Single-unit recordings were made from complex cells in the primary visual cortex of anesthetized adult cats. Each stimulus ensemble consisted of 24,000 grayscale images drawn randomly from a natural movie database (van Hateren and Ruderman, 1998) (Figure 1; see Experimental Procedures). To characterize the nonlinear RF of each neuron, we used a modified STC method to analyze the neuronal response to the natural image

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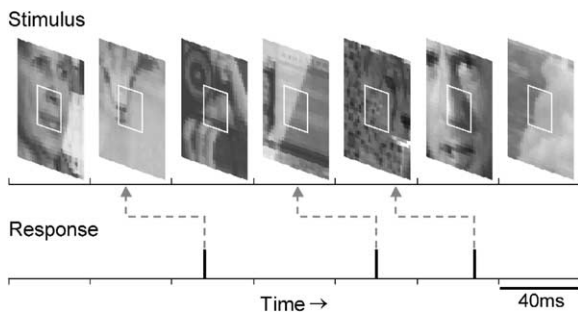


Figure 1. Illustration of Experimental Protocol

(Upper panel) Segment of a natural image ensemble, updated every 41.8 ms. White boxes (12 × 12 pixels) indicate area presented in experiments. (Lower panel) Spike train response. The spike-triggered stimulus ensemble was generated by collecting the natural image preceding each spike by one frame (arrows).

ensemble (Experimental Procedures; also see Figure S1 in the Supplemental Data available with this article online). Briefly, the natural image preceding each spike by one frame (41.8 ms; Figure 1, arrows) was collected to form the spike-triggered stimulus ensemble. This ensemble was normalized by the average power spectrum of natural images to correct for their spatial correlations. The correlation matrix of the spike-triggered ensemble was then computed. Eigenvectors of this matrix with “significant eigenvalues” represent visual features that directly affect the neuronal response. The significant eigenvalues were identified as those that were significantly different from (1) the control eigenvalues calculated on the basis of random spike trains (Figure 2A) and (2) their neighboring eigenvalues (Figure 2B and Experimental Procedures). Figure 2C shows the two significant eigenvectors of an example complex cell. These eigenvectors exhibit oriented ON and OFF subregions, resembling the RFs of simple cells (Hubel and Wiesel, 1962; Jones and Palmer, 1987; DeAngelis et al., 1993). As shown below, the spatial structure of these eigenvectors largely predicts the tuning properties of the neuron. In contrast, the linear RF measured with a modified STA analysis (Theunissen et al., 2000) failed to reveal the spatial structure of the complex cell RF (Figure 2D).

Subunits of Complex Cell RFs

The majority of the complex cells that we analyzed (18/25) had two significant eigenvectors (Figure 2E). For these cells we further examined the spatial profiles of the significant eigenvectors along the axis perpendicular to the preferred orientation and found them to be well approximated by Gabor functions (Figure 3A). In all cases, the Gabor fits for the two significant eigenvectors exhibited similar spatial frequencies, but a phase difference of approximately 90° ($89.5^\circ \pm 5.8^\circ$, mean $\pm$ SD). For a few complex cells (4/25), we found only one significant eigenvector, which also resembled a simple cell RF. For the cells with more than two significant eigenvectors (3/25), the additional eigenvectors exhibited more complex spatial structures. However, simulation studies suggest that these eigenvectors are artifacts

caused by higher-order correlations in natural images (Thomson, 2001) that are not corrected for in the modified STC method (see Figure S2).

While the STC analysis identifies the visual features that contribute to the neuronal response, it does not reveal the quantitative relationship between these features and the response. To address this issue, we computed the contrast-response function of each significant eigenvector (Chichilnisky, 2001; Touryan et al., 2002). First, we measured the contrast of the eigenvector in each natural image as the dot product of the eigenvector and the image (see Experimental Procedures), referred to as “feature contrast.” Second, we measured the instantaneous firing rate in response to each natural image (at a delay of 41.8 ms) and computed the feature contrast-response function as the average firing rate at each feature contrast (Figure 3B). For all cells, the firing rate was found to increase with feature contrast at both positive and negative polarities, consistent with the known polarity invariance of complex cells (Hubel and Wiesel, 1962). To further quantify this function, we fit the data with $r(x) = \beta |x|^\gamma + r_0$ for positive and negative contrasts separately (because the measured contrast-response function is not completely symmetric), where r is the neuronal response, x is feature contrast, and β , γ , and r_0 are free parameters. For these complex cells, the mean exponent (γ) was 1.65 ± 0.35 , comparable to the exponent of contrast-response functions of simple cells (Anzai et al., 1999). Such bimodal contrast-response functions (Figure 3B), together with the spatial phase relationship between the two significant eigenvectors (Figure 3A), are consistent with the energy model (Adelson and Bergen, 1985), which combines a quadrature pair of nonlinear functional subunits to produce an orientation- and spatial frequency-selective but phase-invariant complex cell RF (see Discussion). We thus refer to the significant eigenvectors identified with the modified STC analysis as the subunits of the complex cell RF.

Prediction of Tuning Properties and Responses to Natural Images

The spatial structure of these subunits (Figures 2C and 3A) clearly suggests selectivity for orientation. We next quantified the degree to which the orientation tuning curve predicted by the subunits (derived tuning) matched the tuning curve measured experimentally with drifting gratings (measured tuning). To construct the tuning curve of each subunit, we calculated the dot products of the eigenvector and sinusoidal gratings at a range of orientations at the optimal spatial frequency of the subunit (Figure 4A). These dot products were then converted to the response of the subunit according to the measured contrast-response function (Figure 3B). For cells with two or more significant eigenvectors, we added the tuning curves of the first two eigenvectors to derive the tuning curve of the cell. As shown in Figure 4A, the derived tuning agreed well with the measured tuning of the cell. To quantify this comparison, we fit each tuning curve with a Gaussian function to determine the preferred orientation (peak position) and the tuning bandwidth (width at half height). For the population of cells analyzed, the derived tuning closely

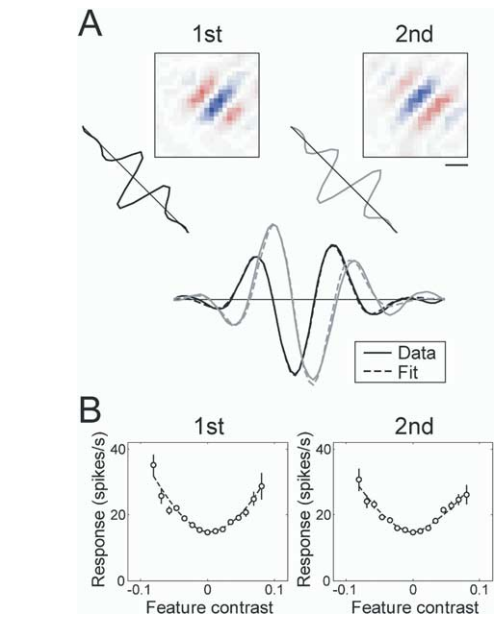
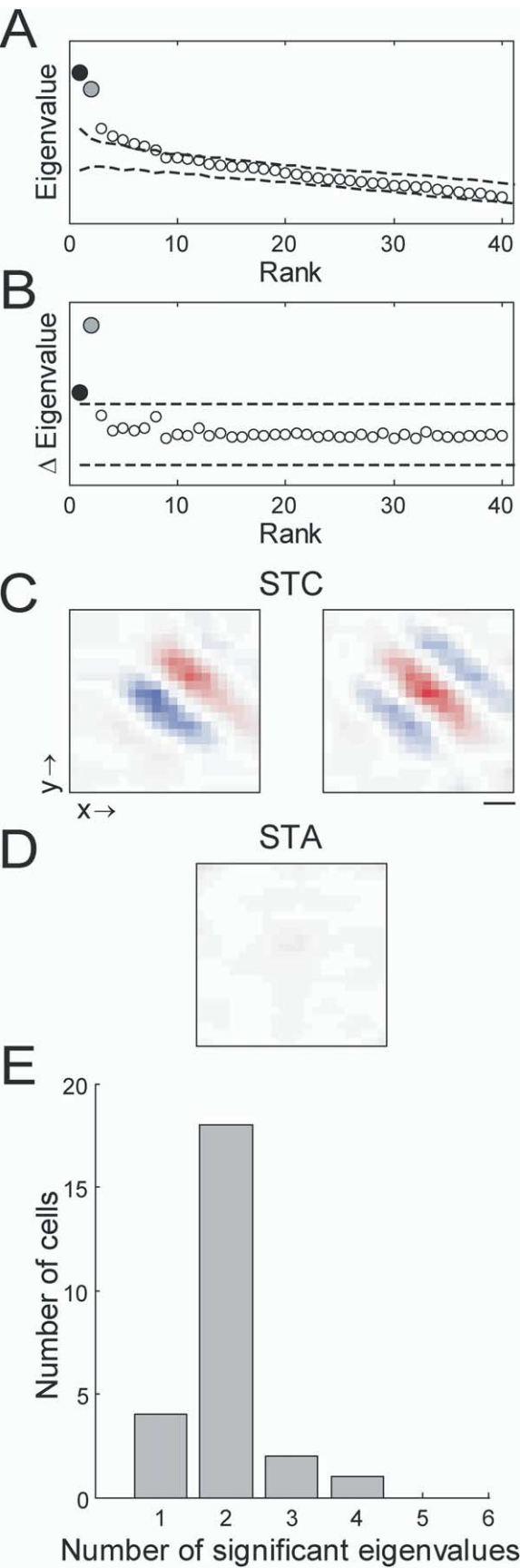


Figure 3. Relationship between the Two Significant Eigenvectors
(A) Significant eigenvectors of an example complex cell. Scale bar, 2°. Solid line, spatial profiles of each eigenvector along the axis perpendicular to the preferred orientation (see Figure 4). Dashed line, Gabor fit. The Gabor fits of the two eigenvectors had a phase difference of 85°. (B) Feature contrast-response functions. Average firing rate is plotted against feature contrast for the two significant eigenvectors shown in (A). Error bar represents $\pm$ SEM. Dashed lines, fits of the data with power functions.

matched the measured tuning in both the preferred orientation (Figure 4B: upper plot, $r = 0.996$; mean absolute difference = 3.6°) and the tuning bandwidth (Figure 4B: lower plot, $r = 0.71$; mean absolute difference = 6.3°). The dot product between the derived and the measured tuning curves for each cell was 0.95 ± 0.04 (mean $\pm$ SD, $n = 25$). Thus, unlike the linear RF (Figure 2D), the significant eigenvectors can accurately predict the selectivity of cortical complex cells for a key stimulus feature.

The significant eigenvectors can also be used to de-

Figure 2. Nonlinear RF of an Example Complex Cell
(A) The 40 largest eigenvalues of the spike-triggered correlation matrix. Dashed lines, control confidence intervals (mean $\pm$ 10 SD) (see Experimental Procedures). (B) Difference between neighboring eigenvalues. Dashed lines, confidence intervals for the difference (mean $\pm$ 10 SD). If a point is found beyond the confidence intervals (i.e., second point, representing difference between the second and third eigenvalues), all the eigenvalues preceding this point are considered significant by this criterion. Filled circles, significant eigenvalues based on criteria in both (A) and (B). (C) The two significant eigenvectors (red, ON; blue, OFF) corresponding to the significant eigenvalues shown in (A) and (B). Scale bar, 2°. (D) The linear RF of the complex cell computed by modified STA. (E) Distribution of the number of significant eigenvectors found for each cell. Cells with no significant eigenvectors were not included.

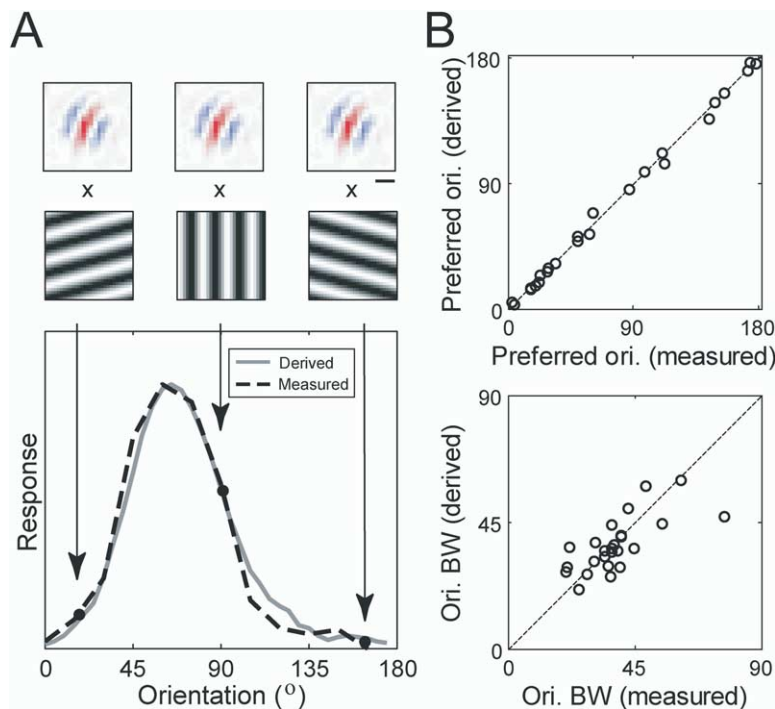


Figure 4. Comparison of Derived and Measured Orientation Tuning Curves

(A) Orientation tuning of a complex cell derived from the significant eigenvectors (solid line) and measured experimentally with drifting gratings (dashed line). Top panel, the first significant eigenvector (upper row) and the grating stimuli (lower row), whose dot product is used to compute the derived orientation tuning curve.

(B) Derived versus measured preferred orientation (upper plot) and tuning bandwidth (lower plot) for all cells ($n = 25$).

rive the spatial frequency tuning of the neuron. The accuracy of the prediction, however, depends on a free parameter used in the modified STC analysis. As described in detail in [Experimental Procedures](#), the stimulus ensemble must be normalized by the average power spectrum of natural images before significant eigenvectors can be identified by STC analysis ([Figure S1](#)). Since the stimulus power at high spatial frequencies is relatively low ([Field, 1987; Dong and Atick, 1995; Simoncelli and Olshausen, 2001](#)), this normalization procedure tends to amplify noise in the high-frequency range ([Theunissen et al., 2000, 2001](#)). To reduce noise amplification, we have limited the normalization to spatial frequencies below a cutoff point (see [Experimental Procedures](#)), which is a critical free parameter in the modified STC analysis. [Figure 5B](#) shows the significant eigenvectors of an example cell computed with different cutoff values (indicated in [Figure 5A](#)). While the orientations of the eigenvectors remained constant, the widths of the ON and OFF subregions and the size of the RF envelope increased systematically with the decreasing cutoff value. To quantify these effects, we derived both the orientation and spatial frequency tuning curves from the eigenvectors computed at different cutoff values and compared them to the experimentally measured tuning curves ([Figures 5C and 5D](#)). We found that while the derived orientation tuning varied only slightly, spatial frequency tuning was more sensitive to the cutoff parameter. At low cutoff values, the eigenvectors appeared less noisy ([Figure 5B](#)) due to reduced noise amplification, but the derived spatial frequency tuning was biased toward low frequencies ([Figure 5D](#)), because signals at spatial frequencies above the cutoff point were attenuated. For the population of cells studied, we found that a cutoff point of 35% provided

a reasonable compromise between the corruption of eigenvectors by high-frequency noise and the bias in derived spatial frequency tuning. Since there is no unique criterion for determining the optimal cutoff value for each cell, the modified STC method is more suited for comparing spatial frequency tuning among a population of neurons than for predicting the exact tuning curve of a particular cell.

Finally, to further validate the significant eigenvectors, we predicted the responses of each neuron to the natural images based on its significant eigenvectors and contrast-response functions, and we computed the correlation coefficient between the predicted and the measured responses. The prediction was made following the same procedure as that used for predicting orientation and spatial frequency tuning, but the significant eigenvectors and their contrast-response functions were recomputed using a fraction of the natural image ensemble (23,000 of the 24,000 frames) separate from that used for the validation (1000 frames). Note that an important factor limiting the correlation coefficient is the variability in the measured responses, due to the small number of repeats (two to four) in our measurements. We estimated the upper bound of the correlation coefficient for each neuron ([Hsu et al., 2004](#); see [Experimental Procedures](#)) and plotted the correlation coefficient between the predicted and the measured responses against this estimated upper bound ([Figure 6A](#)). For most cells the prediction based on the eigenvectors did not reach the upper bound, indicating that the significant eigenvectors did not account for all of the signals in the responses. This may be due to additional subunits that were not identified in the present study (see [Discussion](#)) and/or the inaccuracy of the eigenvectors and contrast-response functions, which

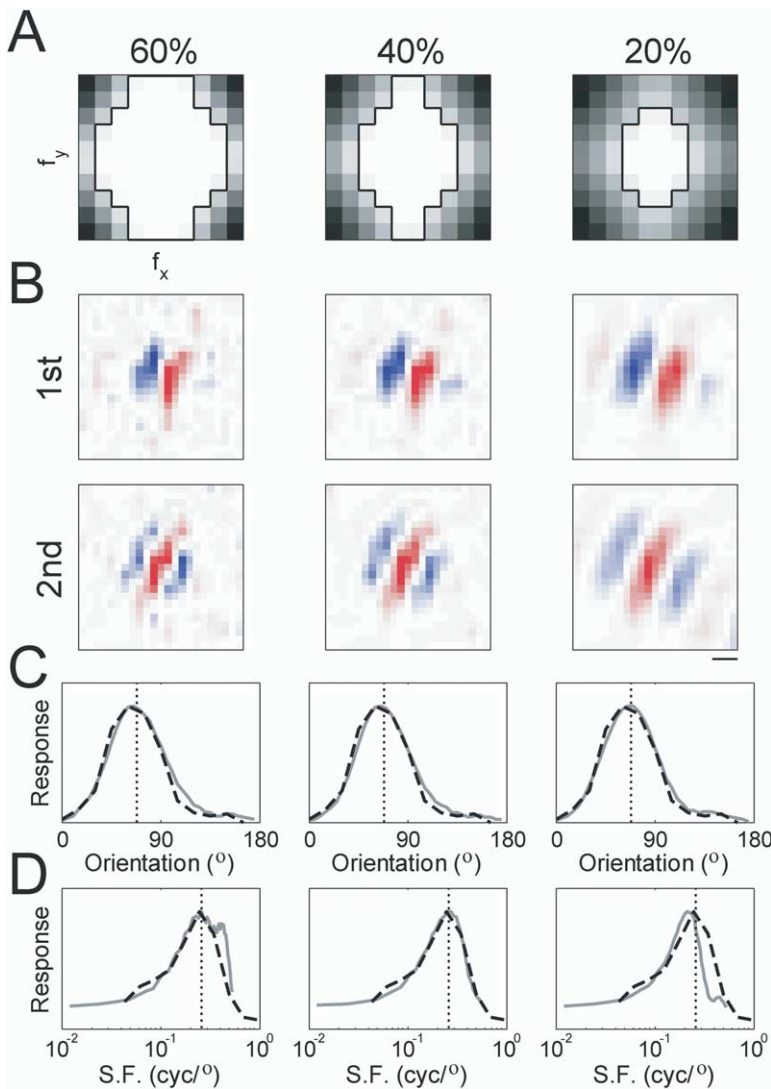


Figure 5. Effect of Normalization Cutoff Point on Estimated RFs

(A) Spatial power spectrum of the stimulus ensemble after normalization with cutoff values of 60%, 40%, and 20%. Black line indicates the approximate range of frequencies over which the stimuli were normalized.

(B) Significant eigenvectors of a complex cell computed with the cutoff values shown in (A). Scale bar, 2° .

(C) Orientation tuning derived from the eigenvectors with different cutoff values (solid line) and experimentally measured tuning (dashed line). Dotted vertical line, preferred orientation measured with 60% cutoff, which is similar to that measured with other cutoff values.

(D) Derived (solid line) and measured (dashed line) spatial frequency tuning. Dotted vertical line, peak spatial frequency with 60% cutoff, which is higher than that measured with lower cutoff values.

were estimated from limited data. Importantly, however, the prediction based on the significant eigenvectors was better than that based on the linear RF computed

with modified STA (Theunissen et al., 2000, 2001; Smyth et al., 2003) for most of the complex cells in this analysis (Figure 6B; for the population, $p < 0.0001$, Wilcoxon signed rank test).

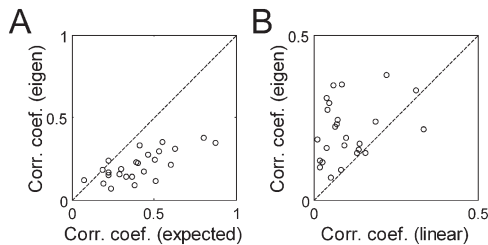


Figure 6. Prediction of Cortical Responses to Natural Images

(A) Correlation coefficient between the predicted and measured responses versus estimated upper bound for the correlation coefficient. Each symbol represents one complex cell.

(B) Correlation coefficients between the predicted and measured responses based on the eigenvectors versus those based on the linear RF.

RFs Measured with Natural Images and Random Stimuli

Given the potential bias in the RFs measured with natural images, it is important to know how these RFs compare to those measured with white-noise stimuli, which are commonly used for RF characterization. Furthermore, given the difference in the statistical properties of natural image and white-noise stimuli, it is conceivable that the cortical neurons are adapted to different states (Maffei et al., 1973; Movshon and Lennie, 1979; David et al., 2004) when processing the two classes of stimuli, resulting in different RF structures. We thus measured the responses of a subset of complex cells ($n = 14$) to both the natural image and random stimulus ensembles (Figure 7A). To allow a fair comparison between these two types of stimuli, the random stimuli

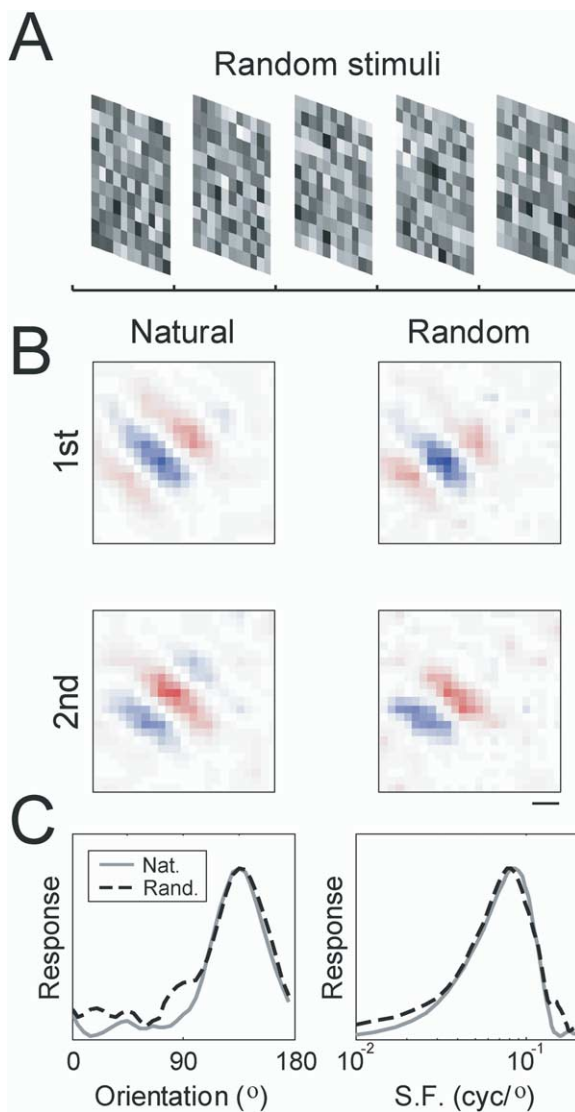


Figure 7. Comparison of the RFs Estimated with Natural Images and Random Stimuli

(A) Random stimuli that are matched to the natural images in both global and feature contrasts (see [Experimental Procedures](#)).

(B) Significant eigenvectors of a complex cell computed from the responses to natural images (left, at 35% cutoff) and random stimuli (right). Scale bar, 2° .

(C) Orientation and spatial frequency tuning derived from the eigenvectors measured with natural (solid line) and random (dashed line) stimuli.

were matched, frame by frame, to the natural images in both the global (root-mean-square) contrast of the image and the feature contrasts of the significant eigenvectors. This was achieved by creating an orthonormal basis set that included the significant eigenvectors of the cell, and for each natural image, a random stimulus was generated by selecting the coefficients of the basis functions such that the global and feature contrasts of the stimulus were matched to those of the natural image (see [Experimental Procedures](#)). Interestingly, although we found no significant difference in either the

mean firing rate ($p > 0.50$) or the burst fraction (defined as the fraction of spikes within groups of two or more spikes with interspike intervals of ≤ 8 ms, based on [De-Busk et al. \[1997\]](#), $p > 0.95$), the signal-to-noise ratio of the responses (measured by the correlation coefficient between repeated trials of the same stimulus) was noticeably higher for the natural images than for the contrast-matched random stimuli (although not statistically significant: $p = 0.057$). Related to this observation, natural images allowed a higher success rate in mapping complex cell RFs: when we applied the STC analysis, significant eigenvectors were found for all the cells from their responses to natural images, whereas with the random stimuli, significant eigenvectors were identified only for 6 of the 14 cells. For these six cells, the spatial structure of the eigenvectors ([Figure 7B](#)) and the derived tuning properties ([Figure 7C](#)) were comparable between the natural images and the random stimuli. For preferred orientation, $r = 0.998$ and the mean absolute difference = 3.3° . For preferred spatial frequency, of course, the result depends on the cutoff parameter: with 35% cutoff, $r = 0.993$ and least-squares regression slope = 0.52; with 60% cutoff, $r = 0.999$ and slope = 0.85. Thus, experiments using random stimuli and natural images reveal similar spatial structure for complex cell RFs, but natural images are considerably more efficient for RF characterization. For these 14 cells, the linear RFs computed from the responses to either random stimuli or natural images (see [Figure 2D](#)) exhibited no clear spatial structure.

Discussion

In the present study, we have identified a small number of functional subunits that are important components of the nonlinear RF of each complex cell. The spatial structure of these subunits resembles simple cell RFs ([Figure 2C](#)), and this functional description of complex cells is consistent with the anatomical model proposed by [Hubel and Wiesel \(1962; also Alonso and Martinez, 1998\)](#). For most cells we found two subunits ([Figure 2E](#)) with a spatial phase difference of $\sim 90^\circ$ ([Figure 3A](#)), and the contrast-response function of each subunit was approximately polarity invariant ([Figure 3B](#)). These findings are consistent with the energy model ([Adelson and Bergen, 1985; Emerson et al., 1992](#)), which can be described as $R = F^\phi(k^\phi \cdot s) + F^{\phi+90^\circ}(k^{\phi+90^\circ} \cdot s)$, where R is the response of the neuron, s is the stimulus, k^ϕ is the RF of a subunit (ϕ represents preferred spatial phase), and F represents the contrast-response function of the subunit. The distinct contribution of the modified STC analysis is that it allows rapid identification of the spatial structure of each subunit from the neuronal responses to natural images (each significant eigenvector corresponds to k^ϕ , $k^{\phi+90^\circ}$, or a linear combination of them [see [Touryan et al., 2002](#)]), leading to a compact description of the RF of each complex cell that directly reveals its feature selectivity. Note, however, that our results do not imply that complex cell RFs are completely characterized by the energy model. First, the $\sim 90^\circ$ phase difference is not surprising for a neuron tuned to a narrow range of spatial frequencies, given the orthogonality constraint for eigenvectors. Second,

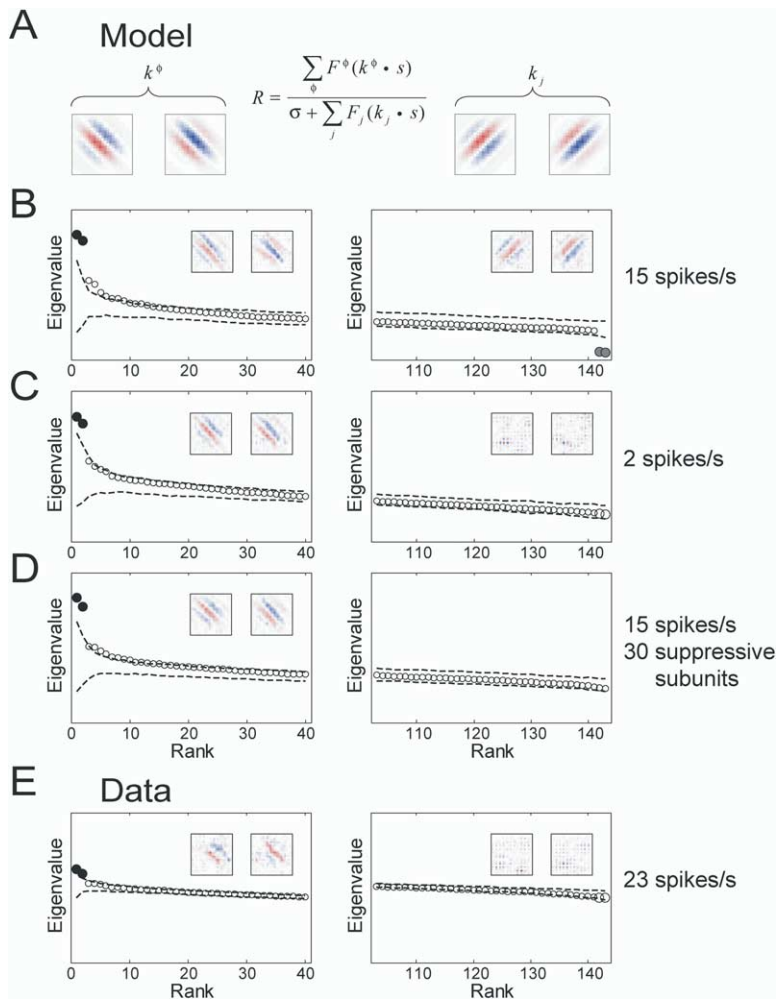


Figure 8. STC Analysis of Suppressive Subunits

(A) A model complex cell with two excitatory (k^{ϕ}) and two suppressive (k_j) subunits as described by the divisive normalization model (see equation).

(B) Excitatory and suppressive eigenvectors computed from a simulated spike train (15 spikes/s) of the model cell in (A) in response to the natural image ensemble. To detect the suppressive subunits, the cutoff point was set at 100%. Dashed lines, confidence interval for the control eigenvalues (mean $\pm$ 10 SD). Note that some of the eigenvalues were beyond the confidence interval, but they were not significant after applying the second criterion (see Figure 2B).

(C) At a lower firing rate (2 spikes/s), STC analysis failed to isolate the suppressive eigenvectors; even the eigenvectors with lowest eigenvalues (larger circles) showed no resemblance to the suppressive subunits of the model.

(D) For a model cell with a firing rate of 15 spikes/s but 30 suppressive subunits, STC analysis failed to isolate the suppressive eigenvectors.

(E) STC analysis of both excitatory and suppressive subunits for a complex cell recorded in cat V1 with a firing rate of 23 spikes/s. Note that, although the firing rate of this cell is higher than that of the model cell shown in (B), no eigenvalues were significantly smaller than control, and the eigenvectors with lowest eigenvalues (larger circles) lacked spatial structure. This was also true for all other complex cells in the present study, suggesting that the suppressive effects in the responses of cat V1 neurons to natural images are distributed over a relatively large number of subunits.

the contrast-response functions of many complex cells are not perfectly symmetric, resulting in quantitative violation of polarity invariance (Spitzer and Hochstein, 1985). Finally, previous studies using grating stimuli have revealed qualitative violations of the energy model (Mechler et al., 2002).

Spike-triggered covariance analysis has been used previously to analyze the responses of cortical complex cells to random-bar stimuli in both the cat (Touryan et al., 2002) and the monkey (Rust et al., 2004), which identified functional subunits that are localized and bandpass along the spatial axis perpendicular to the preferred orientation of the cell. In the study performed in the cat visual cortex (Touryan et al., 2002), the number of significant eigenvectors for each cell was similar to that found in the present study. In the monkey cortex (Rust et al., 2004), however, more significant eigenvectors were found. In addition to the eigenvectors with significantly larger eigenvalues (Figure 2A; excitatory subunits), Rust et al. also found eigenvectors with significantly smaller eigenvalues (suppressive subunits). The difference between the two studies may be attributed to several factors. First, we have used highly stringent criteria for determining significant eigenvectors, which is important for reducing false positives

when this method is used to analyze the responses to natural stimuli (Figure S2). Such stringent criteria may not be necessary for analyzing the neuronal responses to random stimuli, such as those used by Rust et al. Second, there are considerable differences in both the signal-to-noise ratio in the cortical responses and the amount of data used in the analyses (e.g., the number of spikes collected from each cell in the monkey was approximately five times that in the cat), which directly affects the ability of the method to detect subunits. This is illustrated by applying STC to simulated complex cells with various firing rates (Figures 8A–8C). Finally, the number and relative strengths of the additional (excitatory and suppressive) subunits are likely to depend on both the animal species and the stimulus ensemble. In particular, although suppressive effects are known to exist in the neuronal responses in cat V1 (Allman et al., 1985; Bonds, 1989; Walker et al., 2000; Touryan et al., 2002), we did not find any significant suppressive eigenvectors (Figure 8E), perhaps because the effects are distributed over a large number of subunits. As shown in simulations, STC is well suited for detecting a small number of subunits that strongly affect the response (Figure 8B), but its sensitivity decreases if the suppressive effects are distributed over a large number

of subunits (Figure 8D). Thus, our results do not necessarily contradict the findings reported by Rust et al. (2004).

In addition to STC, complex cell RFs have been analyzed recently using a phase-separated Fourier model (David et al., 2004). This method employs a nonlinear transformation of the stimulus followed by modified STA, and it provides a description of the complex cell RF in the frequency domain. In contrast to our finding that the RFs measured with natural images and with random stimuli were similar, the above study showed that the spatiotemporal RFs depend significantly on the statistics of visual stimuli. This difference may be largely explained by the fact that the strong temporal correlation in natural vision movies, shown to be the principal cause for the stimulus dependence of both spatial and temporal RF properties (David et al., 2004), was virtually absent in the natural image sequences used in our study (see below). In addition, David et al. compared the RFs measured with natural stimuli and dynamic gratings, while in the present study we compared RF measurements based on natural images and broadband random stimuli.

Several methods have proved useful for estimating visual cortical RFs from the responses to complex stimuli, including iterative/recursive methods (e.g., least-squares algorithms [Ringach et al., 2002], artificial neural networks [Lehky et al., 1992; Lau et al., 2002; Prenger et al., 2004], and information maximization [Sharpee et al., 2004]) and methods involving statistical characterization of the spike-triggered stimulus ensemble (e.g., STA [Jones and Palmer, 1987; DeAngelis et al., 1993; Reid et al., 1997; Smyth et al., 2003] and STC [Touryan et al., 2002; Rust et al., 2004]). In general, the spike-triggered methods are much faster (on the order of seconds), allowing online characterization of the neuronal RFs during experiments (Touryan et al., 2002), but it is more difficult to apply them to nonwhite stimuli (e.g., natural stimuli) due to the high degree of correlation in these stimuli. The second-order correlations can be corrected for by normalizing the stimulus ensemble by its power spectrum (Theunissen et al., 2000, 2001; Smyth et al., 2003; David et al., 2004), but this procedure often results in noise amplification at the frequencies with low signal power (e.g., high-frequency range). As shown in the present study, limiting normalization to the spatial frequencies below a cutoff point reduces the random noise in the eigenvectors but causes a systematic bias toward low frequencies (Figure 5), a tradeoff also necessary in modified STA methods (Theunissen et al., 2000, 2001; Smyth et al., 2003; David et al., 2004). The optimal value of the cutoff parameter, of course, depends on the relative importance of these two factors to the particular problem being addressed. Furthermore, since noise in the estimated RF directly depends on the amount of data used in the analysis, as the amount of data increases, the cutoff point can be raised accordingly to reduce the bias in the estimation.

In addition to spatial correlations, natural movies also exhibit a high degree of temporal correlations (Dong and Atick, 1995). The stimuli used in the present study, however, consisted of a random sequence of natural images with little temporal correlation. These stimuli

were used for two reasons. First, they consist of a much broader range of images than natural movies of the same length, which allows a less biased estimation of the RFs. Second, the lack of temporal correlation significantly simplifies RF estimation, since stimulus normalization (see above) can be limited to the spatial domain. Thus, given that the purpose of this study is to examine the spatial structure of complex cell RFs, the natural image ensembles used here offer important experimental and computational advantages. The spatial RFs measured with the present method can serve as a basis for future studies using natural time-varying images.

Natural stimuli have been used with an increasing frequency to measure neuronal RF properties (Theunissen et al., 2000, 2001; Ringach et al., 2002; Smyth et al., 2003; David et al., 2004). The present study extends the use of natural images to the estimation of nonlinear RFs in the visual cortex. Interestingly, we found natural images to be more effective than random stimuli in mapping complex cell RFs. Compared to neurons in the retina and thalamus, cortical neurons are likely to be progressively more selective for visual features in natural scenes (Tanaka, 1996; Kanwisher et al., 1997). Thus, the use of natural images in conjunction with nonlinear analyses may constitute a powerful approach for understanding the RF properties of these high-level neurons.

Experimental Procedures

Recording

Animal use procedures were as previously described (Touryan et al., 2002) and were approved by the Animal Care and Use Committee at the University of California, Berkeley. Single-unit recordings were made in area 17 of adult cats (2–6.5 kg) using tungsten electrodes (A-M Systems, Carlsborg, WA); cells were sampled at all laminar locations. Unit isolation was based on cluster analysis of waveforms and the presence of a refractory period determined from the autocorrelograms. Cells were excluded from the analysis if their mean firing rates were <1 spike/s in response to natural images, or if their responses were not significantly correlated between different repeats of the same stimulus. Cells were classified as simple if their RFs had clear ON and OFF subregions (Hubel and Wiesel, 1962) and if the ratio of the first harmonic to the DC component of the response to an optimally oriented drifting grating was >1 (Skottun et al., 1991). All cells included in this study were complex cells.

Visual Stimulation

Visual stimuli were generated with a PC and presented with a Barco monitor (size 40 × 30 cm, refresh rate 120 Hz, maximum luminance 80 cd m⁻²). Luminance nonlinearities were corrected through software. Two types of stimulus ensembles were used: natural images and random stimuli.

Natural Image Ensemble

Images were selected at random from a database consisting of a variety of digitized movies (van Hateren and Ruderman, 1998), and the center patch (12 × 12 pixels) of each image was retained. To maximize the diversity of images, we measured the similarity between each pair of stimuli S_i and S_j in the ensemble by their dot product

$$\sum_{y=1}^{12} \sum_{x=1}^{12} S_i(x, y) S_j(x, y) / \sqrt{\left(\sum_{y=1}^{12} \sum_{x=1}^{12} [S_i(x, y)]^2 \right) \left(\sum_{y=1}^{12} \sum_{x=1}^{12} [S_j(x, y)]^2 \right)}$$

for similar images (dot product > 0.95), either S_i or S_j was excluded. Each ensemble consisted of 24,000 images; three distinct natural ensembles were used in this study. Unlike natural movies, these

images were presented in a random sequence; the absence of temporal correlation facilitated the spike-triggered covariance analysis (see below). However, since natural images are highly variable in their mean luminance and global contrast (measured by root-mean-square contrast

$$\sqrt{\frac{1}{144} \sum_{y=1}^{12} \sum_{x=1}^{12} [S(x, y)]^2},$$

where $-1 \leq S(x, y) \leq 1$, and 1 and -1 represent highest and lowest luminance of the monitor, respectively), such stimuli may invoke contrast adaptation (Maffei et al., 1973; Movshon and Lennie, 1979; Ohzawa et al., 1982) that could confound the interpretation of the results. To control for this effect, we removed the mean and scaled each image such that all frames had the same global contrast, which was 0.32.

Random Stimulus Ensemble

The random stimulus ensemble was created specifically for each cell to match the natural image ensemble in both the global and feature contrasts (see below) to allow a fair comparison between the two types of stimuli in RF measurement. First, we created an orthonormal basis set (V_i , $i = 1, 2, \dots, 144$) that included the significant eigenvectors of the cell (e.g., for a neuron with two significant eigenvectors, V_1 and V_2 represent the eigenvectors, and $V_3, \dots, V_{144}$ are arbitrary aside from the requirement of orthonormality). Then, for each frame in the natural image ensemble, we generated a corresponding "random" stimulus as

$$S(x, y) = \sqrt{144} (c_1 V_1(x, y) + c_2 V_2(x, y) + \sum_{i=3}^{144} c_i V_i(x, y)).$$

The coefficients $\{c_i\}$ were chosen such that (1) c_1 and c_2 were the contrasts of the first and second significant eigenvectors in the natural image ensemble, which ensured that the random stimulus was matched to the natural image for feature contrast and (2)

$$\sum_{i=1}^{144} c_i^2 = \frac{1}{144} \sum_{y=1}^{12} \sum_{x=1}^{12} [S(x, y)]^2,$$

where S is the natural image. This ensured that the global stimulus contrast (equal to

$$\sqrt{\sum_{i=1}^{144} c_i^2})$$

for an orthonormal basis set) was also matched to that of the natural image. This process was repeated for all the frames in the natural image ensemble. Note that although the stimuli that satisfied constraints (1) and (2) are, strictly speaking, not random, they exhibit no clear spatial structure because of the randomness in choosing most of the basis functions ($V_3, \dots, V_{144}$) and their coefficients ($c_3, \dots, c_{144}$). The spatial power spectra of these stimuli are essentially flat, similar to white-noise stimuli.

Both the natural and the random stimulus ensembles were updated every five frames, corresponding to an effective frame rate of 24 Hz (41.8 ms/frame, 16.7 min/ensemble). For each cell, the stimuli were presented in an area slightly larger than the classical RF of the cell estimated by hand mapping.

Spike-Triggered Covariance Analysis

Unlike the spike-triggered average, which is the mean of the spike-triggered stimulus ensemble (Jones and Palmer, 1987; DeAngelis et al., 1993; Reid et al., 1997; Ringach et al., 1997), the spike-triggered covariance analysis identifies a set of visual features (represented by eigenvectors of the spike-triggered correlation matrix) that account for different amounts of variance (corresponding eigenvalues) in the ensemble. This method is closely related to Wiener kernel analysis (Wiener, 1958; Gaska et al., 1994), and it computes the eigenvalues and eigenvectors of the second-order kernel. Here, we first provide a brief description of the original STC method applicable to white-noise stimuli and then discuss the modified method for nonwhite stimuli.

STC Analysis for White-Noise Stimuli

For white-noise stimuli with a zero mean, the spike-triggered correlation matrix $[C_{m,n}]$ is computed as

$$C_{m,n} = \frac{1}{N} \sum_{i=1}^N S_m(i) S_n(i),$$

where $S_m(i)$ and $S_n(i)$ are the luminance values of the m th and n th pixels in the stimulus preceding the i th spike, respectively, and N is the total number of spikes in the response. Eigenvalues and eigenvectors of this spike-triggered correlation matrix are then computed to identify the significant eigenvectors (Touryan et al., 2002) (see below).

Modified STC Analysis for Natural Images

To analyze the responses to nonwhite stimuli, it is necessary to correct for the stimulus correlations by "whitening" (or "normalizing") each stimulus in the ensemble:

$$S_w = SU \begin{pmatrix} \frac{1}{\sqrt{\lambda_1}} & & 0 \\ & \ddots & \\ 0 & & \frac{1}{\sqrt{\lambda_n}} \end{pmatrix},$$

where S is a vector representation of the stimulus (e.g., luminance in each pixel), U is a matrix containing the eigenvectors of the correlation matrix of S , and $\lambda_1, \dots, \lambda_n$ are the corresponding eigenvalues. As a result, S_w represents the stimulus in the whitened space. Spike-triggered covariance analysis for white-noise stimuli can then be applied to the ensemble $\{S_w\}$ to identify the significant eigenvectors (V_w) in the whitened space (see below). The significant eigenvectors in terms of luminance value in each pixel and frame can then be computed as

$$V = V_w^T \begin{pmatrix} \frac{1}{\sqrt{\lambda_1}} & & 0 \\ & \ddots & \\ 0 & & \frac{1}{\sqrt{\lambda_n}} \end{pmatrix} U^{-1},$$

which correspond to the subunit RFs. To illustrate the importance of the normalization procedure in subunit identification, we applied the STC analysis, with and without normalization, to the simulated spike train of a model complex cell in response to the natural image ensemble. As shown in Figure S1, normalization is crucial for the successful identification of the model cell subunits.

Practically, however, the above method needs to be further modified. The eigenvectors (U) and eigenvalues ($\lambda_1, \dots, \lambda_n$) of the stimulus correlation matrix (S) approximate the Fourier components and the power spectrum of natural images, respectively. Since the spatial power spectrum of natural scenes is approximately $1/k^2$ (where k is spatial frequency) (Field, 1987), the eigenvalues (λ) corresponding to eigenvectors with high spatial frequencies can be very small. The normalization step (dividing by a small number $\sqrt{\lambda}$) will thus result in noise amplification. To solve this problem, we chose a cutoff point such that normalization is only performed for eigenvectors above this cutoff point (expressed in terms of the percentage of the total number of eigenvectors, e.g., 35% cutoff indicates that 50 of the 144 eigenvectors were normalized). Since the high-frequency components above the cutoff point were not normalized, this procedure results in a bias of the estimated complex cell RF toward lower spatial frequencies (Figures 5B and 5D).

Identification of Significant Eigenvectors

To determine the "significant eigenvectors," we performed the following analysis. First, we identified which eigenvectors of the spike-triggered stimulus ensemble had eigenvalues significantly larger than the control eigenvalues (Figure 2A). The control eigenvalues are defined as the eigenvalues of spike-triggered stimulus ensembles based on random spike trains, each of which had the same number of spikes as the recorded spike train, but with random spike timing. The confidence interval for the control eigenvalues was computed using 50 random spike trains (mean ± 10 SD). Second, we identified eigenvalues that stood out from the rest of the eigenvalues of the spike-triggered ensemble by calculating

the difference between neighboring eigenvalues (Figure 2B). In this case, the control confidence interval was set as the mean ± 10 SD of the difference values, but only for the eigenvalues that did not pass the first criterion (points between the dashed lines in Figure 2A). If a point is found beyond the confidence intervals, all the eigenvalues preceding this point are considered significant by this criterion. Note that, for STC analysis with white-noise stimuli, only the first criterion is necessary for eigenvector identification (Touryan et al., 2002; Rust et al., 2004). The second criterion was incorporated in the present study because the higher-order correlations in natural images (Thomson, 2001) (not corrected for in the modified STC method) render the first criterion less reliable, leading to a significant number of false positives (Figure S2). Our simulation studies showed that adding the second criterion significantly improves the reliability of the identification procedure.

Although the above method can be used to estimate spatiotemporal eigenvectors, in the present study we focused on the spatial domain in order to improve the signal-to-noise ratio of the estimate. The analysis was performed only at the frame 41.8 ms before each spike, which was determined in preliminary studies to be the optimal temporal delay. For a subset of the cells with high signal-to-noise ratio in their responses, we found that performing the full spatiotemporal analysis did not significantly change the spatial profile of the significant eigenvectors. For each cell, we used the responses to two to four repeats of a natural image ensemble (consisting of 24,000 different images) to compute the significant eigenvectors.

Feature Contrast-Response Function

The contrast of the visual feature (significant eigenvector) $V(x, y)$ in stimulus $S(x, y)$ is measured as the dot product of V and S :

$$\frac{1}{\sqrt{144}} \sum_{y=1}^{12} \sum_{x=1}^{12} V(x, y) S(x, y);$$

$$V(x, y) \text{ satisfies } \sum_{y=1}^{12} \sum_{x=1}^{12} V(x, y) V(x, y) = 1.$$

This definition ensures that the feature contrast in each stimulus is bound between -1 and 1 . For each feature, the contrast-response function was measured from the neuronal responses to two to four repeats of the ensemble.

Estimation of Upper Bound

The upper bound of the correlation coefficient between the predicted and the measured responses (averaged over M repeats) is estimated as the expected correlation coefficient between the actual firing rate (i.e., that measured over infinite number of repeats) and the firing rate averaged over M repeats (Hsu et al., 2004; M is an even number):

$$r_{AR_M}^2 = \frac{2}{1 + \sqrt{1/r_{R_1, M/2}^2 R_{2, M/2}}},$$

where A is the actual firing rate, $\bar{R}_M$ is the average firing rate over M repeats, $\bar{R}_{1, M/2}$ is the average of $M/2$ repeats, and $\bar{R}_{2, M/2}$ is the average of the other $M/2$ repeats (Hsu et al., 2004). For each cell with >2 repeats, we computed $r_{AR_M}^2$ for each possible pair of $\bar{R}_{1, M/2}$ and $\bar{R}_{2, M/2}$ and used the average result as the estimated upper bound for the cell.

Supplemental Data

Supplemental Data include two figures and can be found with this article online at <http://www.neuron.org/cgi/content/full/45/5/781/DC1/>.

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